



NINHYDRIN



The Molecule That Makes Invisible Evidence Visible

1 IDENTITY

IUPAC Name:

2,2-dihydroxyindene-1,3-dione

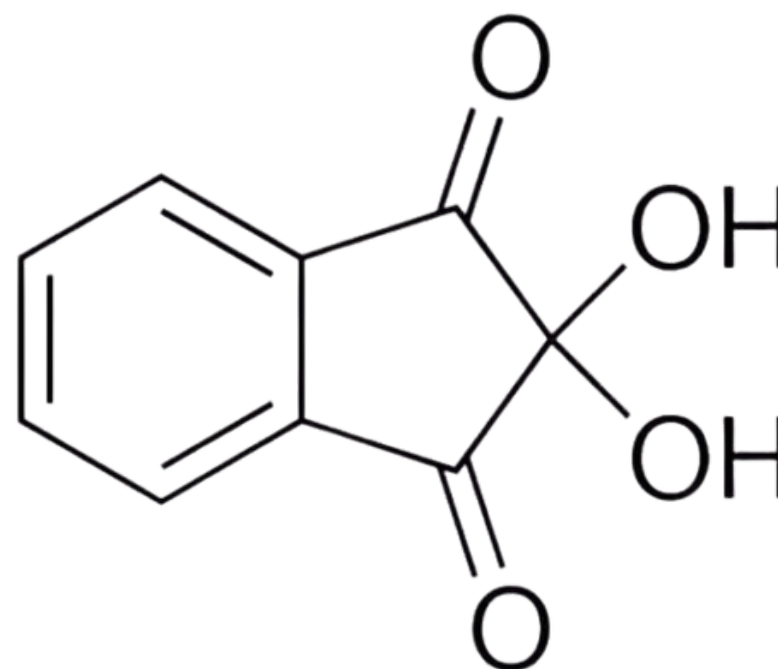
Formula: $C_9H_6O_4$

Molar Mass: 178.14 g/mol

Specialty:

Detecting amino acids and latent fingerprints

2 STRUCTURE & IDENTITY



Functional Group:

- Carbonyl groups
- Hydroxyl groups

Notable Feature:

- Fused ring structure

COOL CHEMISTRY FACT

Ninhydrin reacts strongly with amino acids. This reaction is the key to making invisible evidence visible.

3 IMPORTANT PROPERTIES

- White to pale yellow crystalline powder
- Reacts strongly with amino acids
- Produces a deep purple color
- Sensitive to light
- Can decompose when strongly heated

4 WHAT MAKES IT AMAZING?

IT REVEALS THE INVISIBLE!

Invisible
fingerprint
residue

Reacts with
ninhydrin

Forms
Ruhemann's
Purple

Fingerprint
becomes
visible!

Chemical reactions can turn hidden molecules into visible evidence

5 USES & VALUE

- Detecting latent fingerprints
- Detecting amino acids
- Biochemical and chemical analysis



- Laboratory testing
- Forensic investigations

6 RISK & SAFETY



- May irritate skin and eyes
- May irritate the respiratory system

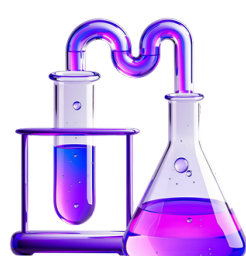
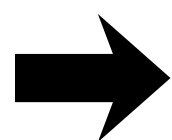
- Use gloves, goggles, and a lab coat
- Work in a well-ventilated area

7 CHEMICAL ENGINEERING CONNECTION



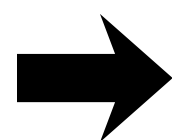
REACTION ENGINEERING

Control conditions
for effective color
development



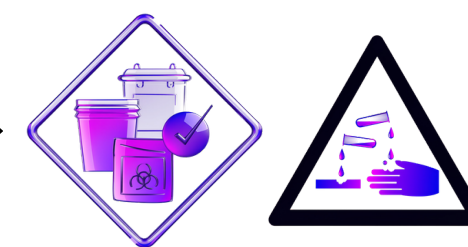
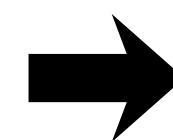
FORMULATION

Develop effective
reagent mixtures



PROCESS OPTIMIZATION

Improve reaction
time and consistency



SAFETY & WASTE

Minimize exposure
and properly treat
chemical waste

8 REFERENCE



- National Center for Biotechnology Information. (n.d.). PubChem compound summary for CID 10236, ninhydrin. PubChem.
- National Institute of Justice. (2011). The fingerprint sourcebook. U.S. Department of Justice.